



HACKATHON SUPPORTING MATERIAL

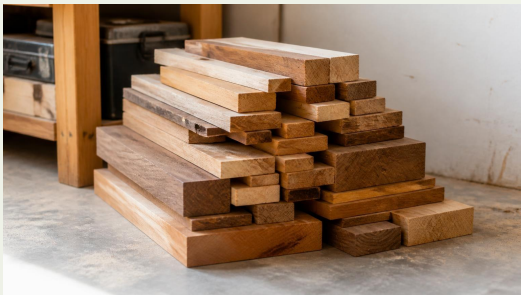
CircularCut

Use existing timber first

EarthSync / Shah, Kai, Tasfia, Jaycee and Shelly



Finding surplus is only half the problem



Surplus exists

Workshops have usable timber offcuts while others purchase new timber.



90 x 45 Treated Pine Outdoor Timber Structural H3 Set Lengths

On Sale! In Stock

Length : (Required)

Choose Options

Width: 90mm

Thickness: 45mm

Material: Outdoor H3 Treated Pine

Grade: Certified Structural

Sold as: Per Length

\$30.78 - \$51.30

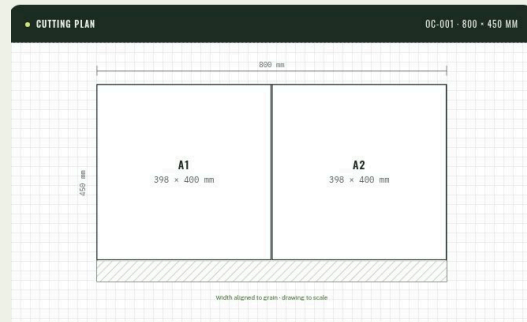
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Add +



A listing is not a fit

A listing does not establish whether an offcut works for a specific order.

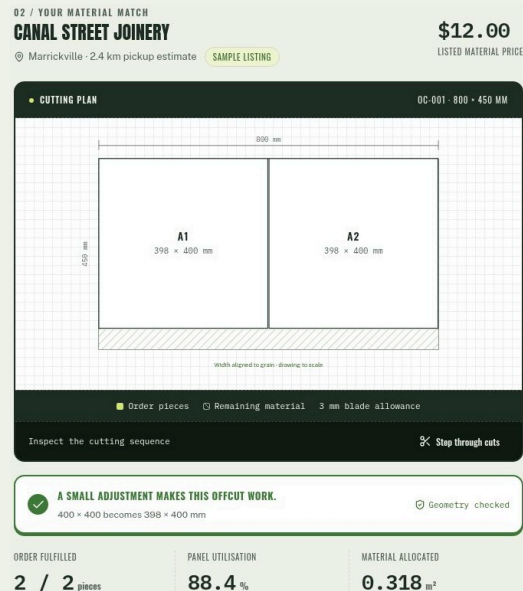


Physical fit decides

Dimensions, cutting losses and saw-blade kerf determine physical fit.

A matching platform that checks the cut

- Buyers specify material, thickness, dimensions, quantity and acceptable minimum dimensions.
- CircularCut matches available offcuts, checks geometry and accounts for kerf.
- Buyers review a visual cutting plan, utilisation and remaining material before reserving.



ORDER → MATERIAL MATCH → GEOMETRY CHECK → CUTTING PLAN →
RESERVATION → REUSE TRACKING → IMPACT EVIDENCE

Plan the cut before reserving the timber

Choose an offcut and enter requested panel dimensions, quantities and minimum acceptable sizes.

Set saw kerf and whether rotation is allowed; the plan recalculates as inputs change.

Compare visual fit and no-fit results, remaining material and yield before reserving a suitable offcut.

Validated demo: two 400 x 400 mm panels.
The buyer approves 398 x 400 mm panels within the stated minimum sizes.

MATERIAL FIRST. POSSIBILITY NEXT.

FIND AN OFFCUT FOR YOUR NEXT ORDER.

Give good material a second job, with a cutting plan that fits.

mm · AUD

01 YOUR ORDER

PROJECT NAME
Two display panels

MATERIAL
Birch plywood

THICKNESS (MM)
18
Clean panels
Non-structural use

PIECES TO MAKE 2 pieces

A Display panel
WIDTH A (MM) × HEIGHT A (MM) QTY A
400 × 400 2
Minimum accepted
MIN WIDTH A MIN HEIGHT A
390 × 400 mm

+ Add part type

ALLOW A SMALL SIZE ADJUSTMENT ☒
Only inside the minimum sizes above.

CUTTING SETTINGS 3 mm blade
One rectangular offcut per order. Whole-millimetre adjustments, up to 20 mm per dimension.

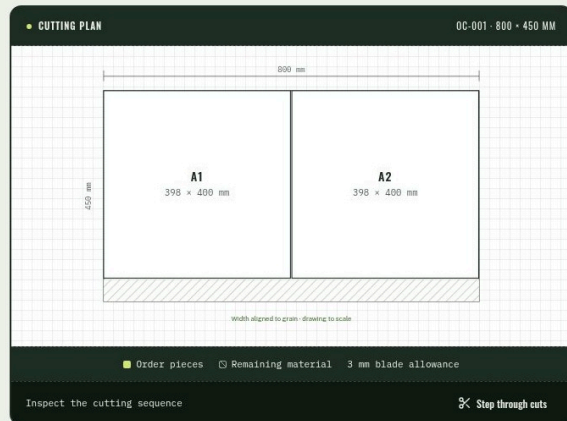
02 / YOUR MATERIAL MATCH CANAL STREET JOINERY

📍 Marrickville · 2.4 km pickup estimate

[SAMPLE LISTING](#)

\$12.00

LISTED MATERIAL PRICE



A SMALL ADJUSTMENT MAKES THIS OFFCUT WORK.

400 × 400 becomes 398 × 400 mm



Geometry checked

ORDER FULFILLED

2 / 2 pieces

PANEL UTILISATION

88.4 %

MATERIAL ALLOCATED

0.318 m²



REVIEW & RESERVE THIS OFFCUT



PRINT CUTTING PLAN

The seller receives an in-app update. Agree pickup in the reservation conversation; pickup and cutting costs are additional.

VALIDATED DEMONSTRATION

A small approved adjustment makes an offcut usable

REQUESTED ARRANGEMENT

Two requested panels from 800 × 450 mm stock

400 × 400 mm

400 × 400 mm

$400 + 3 + 400 = 803$ mm

3 mm wider than the 800 mm offcut.

ADJUSTED ARRANGEMENT

If the buyer accepts a minimum width of 390 mm, the prototype proposes:

398 × 400 mm

398 × 400 mm

$398 + 3 + 398 = 799$ mm

Fits within the buyer-approved dimensions.

MARKETPLACE

GOOD MATERIAL, WAITING FOR ITS NEXT JOB.

Known materials, seller details and pickup arrangements make a useful match possible.

+ LIST AN OFFCUT

Q Search material, workshop or suburb

All materials

7 offcuts

OC-001 SAMPLE



800 × 450 × 18 mm

BIRCH PLYWOOD

\$12.00

Canal Street Joinery
Marrickville - 2.4 km

AVAILABLE REPORTED CLEAN

► SELLER, PICKUP & DISPOSAL BASELINE

PLAN WITH THIS OFFCUT

OC-002 SAMPLE



900 × 600 × 18 mm

BIRCH PLYWOOD

\$22.00

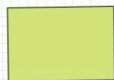
Second Bench Studio
Tempe - 4.8 km

AVAILABLE REPORTED CLEAN

► SELLER, PICKUP & DISPOSAL BASELINE

PLAN WITH THIS OFFCUT

OC-003 SAMPLE



650 × 450 × 18 mm

BIRCH PLYWOOD

\$9.00

Neighbourhood Makers
Dulwich Hill - 1.1 km

AVAILABLE REPORTED CLEAN

► SELLER, PICKUP & DISPOSAL BASELINE

PLAN WITH THIS OFFCUT

OC-004 SAMPLE



1000 × 600 × 12 mm

OC-005 SAMPLE



1220 × 610 × 18 mm

OC-006 SAMPLE



1200 × 800 × 18 mm

Compare available offcuts before reserving

The marketplace shows material, dimensions, thickness, condition and availability.

Alternative offcuts can be compared against buyer requirements.

Prototype listings and quantities are demonstration data, not verified exchanges.

Useful matching starts with a measured listing

The seller records material, dimensions, thickness, condition and defect notes.

Pickup details and the reported alternative destination help the buyer assess collection.

Seller reports remain unverified.
Unknown disposal destinations stay unknown.

GIVE AN OFFCUT ITS NEXT OPPORTUNITY.
Record a clean rectangular panel with known dimensions.

MATERIAL
Birch plywood

WIDTH (MM) 800 **HEIGHT (MM)** 450 **THICKNESS (MM)** 18

WORKSHOP My workshop **SUBURB**

LISTED PRICE (A\$) 0 **PICKUP DISTANCE ESTIMATE (KM)** 0

Clean, known material Grain along width

SELLER & PICKUP DETAILS

DEFECTS, INSPECTION AND PREVIOUS-USE NOTES
Describe edge damage, surface marks or inspection needed

PICKUP AVAILABILITY For example, weekdays 10 am–3 pm **PICKUP PLACE OR ACCESS NOTES** Collection area, loading access or st

SELLER CONTACT OR PREFERRED CONTACT METHOD
Contact details or use the pickup conversation

ALTERNATIVE DESTINATION

Measure completed reuse

One completed 398 x 400 mm panel records 0.159 m² of reused material.

Reservation and collection alone add no completed reuse. Weight stays blank unless the maker reports a measurement.

Demonstration data from fictional stock.
Disposal baselines and outcomes are self-reported.
No measured carbon or methane savings claimed.

MEASURE WHAT ACTUALLY HAPPENS

A CLEAR TRAIL FROM OFFCUT TO OUTCOME.

Planning shows potential. Recorded reuse shows what the maker reports completing.

● SAMPLE RECORDS

⚠ These are demonstration outcomes from fictional stock. They are not real-world impact.

AREA RECORDED REUSED

0.159 m²

Only pieces selected at completion

PIECES GIVEN A SECOND USE

1

Across 1 completed order

WEIGHT REPORTED WIELDED

—

0 of 1 records include weight

NEW PURCHASES REPLACED

0 orders

Maker reported counterfactual

THE ALTERNATIVE DESTINATION

REUSE WITH A RECORDED DISPOSAL BASELINE.

1 of 1 completed sample records include an alternative destination and the seller's explanation, 0 remain unknown.

0.159 m²

Recorded reused area with a reported landfill baseline

Only completed pieces count. The disposal route is seller-reported and has not been independently verified. This is not a methane or CO₂ estimate.

BUILD FOR 2025

GREEN INDUSTRIALIZATION

Help small producers use existing materials in their next order. The barrier is practical: finding suitable stock and checking whether the required pieces can actually be cut from it.

📦 Primary: circular material use

🗑 Secondary: keeping useful material out of waste streams

Reuse is recorded separately from reclamation. Landfill diversion and emissions savings require evidence about the alternative disposal route and a suitable assessment method.

WHAT THE CUTTING ENGINE CHECKS

- ✓ Material and thickness match the order
- ✓ All pieces stay inside the trimmed boundary
- ✓ Pieces have the specified blade clearance
- ✓ Sizes stay within the buyer's allowed range
- ✓ Rotation follows the order setting
- ✓ Pieces, remnants, blade loss and trim reconcile

The engine searches for quality fit marks and shows which millimeters clear reductions. It returns a valid plan when found; it does not guarantee the global optimum or prove that every rejected order is impossible.

Make circular production practical at workshop level

Primary priority

Green Industrialization

Use existing resources more efficiently and potentially reduce demand for newly produced timber.

Secondary: waste prevention

Zero Waste & Methane Reduction

Keep usable offcuts in productive use rather than discarding them. Do not assume every discarded offcut would generate methane; claims depend on verified disposal pathways.

Build for 2035: help small producers use existing material in their next order.
Circular material use is the primary climate contribution.

The next step is real-world testing

- 1 Proposed pilot: five local workshops and twenty real orders, using clean, measured offcuts.
- 2 Have makers review cutting plans and compare matching time with usual procurement.
- 3 Track collection, completed pieces, reused area or weighed mass, and reported new purchases replaced.

Proposed pilot. No physical maker trial completed yet.

AI disclosure: ChatGPT/OpenAI assisted the build and writing.
Claude assisted prototype/design work. Gemini generated animation.

Narration recorded by the team. Core matching uses deterministic calculations.

**What
existing
material
can be
used
first?**